data about where the livestock is. Besides aiding in farming this could help scientists study components of animal behaviour. Other possibilities of using IoT to improve agricultural and farming methods is to fit sensor systems in various strategic points across the farm like gates and tanks. The farmer can monitor water levels, leakages and structural damage and also look out for intruders. With automation systems industry coming up with constant innovations it might be possible to use IoT to do remote farming. Imagine the sight of a farmer sitting in the comfort of his home running tractors, sprinklers and what not from his phone. The Kirby farm in Australia is a working model of a smart farm installed with many sensors to gain data about soil moisture, salinity and temperature. The combination of sensors and livestock tagging allows the farmer to know which areas in the farm are better for the livestock and which are prone to harbour parasites. Similar sensors can also be fitted near river beds to get data about erosion and soil nature besides acting as a flood warning system. For Greenhouse based farms, these sensors can be installed inside greenhouses to enable continuous monitoring of temperature and humidity. Large farms with associated processing, storage and transportation facilities can benefit greatly from IoT technology. Produce can be tracked, processing stages monitored and controlled remotely, and various stages of supply chain can be seamlessly managed.

A company called Monsanto is using IoT in a combination of ways described above. The company has an application and program that helps the farmer single out favourable areas for growing crops and then enable to monitor the stages of planting. The company uses data of the produce to help its trademark program make better predictions. IoT has also been successfully implemented in creating a pest control system. The farmer can use visuals from installed cameras and data from sensors to help him decide when to release insect pheromones for pest control. The company that has pioneered this (SemiosBIO) claims that pesticide usage is reduced significantly by intelligent use of IoT.

The potential of IoT in enhancing the quality of living in cities and rural areas is proven. However with the kind of problems IoT is trying to target and solve, there are many challenges and hurdles, the biggest of which is arguably scalability and robustness. A sensor based technology can perform very well in a controlled setting but out there in a free environment , delicate technology can fail quite easy. Can an efficient design be adapted to be an efficient and robust design that can perform in an uncontrolled